Fragrance Data Analysis Report

Summary:

The dataset contains 24,063 perfumes with 1,671 distinct notes. The analysis shows that scent profile carries real signal for style and gender label, but only modest signal for how well a perfume is rated.

- Style: Style is learnable, a model using only notes and accords predicts the gender label with **71.9%** accuracy, and k-means finds ten interpretable scent families.
- Rating: Ranking is hard to predict, the best model explains **16.1%** of rating variance, which fits the subjective nature of taste.
- Raw Rating: Ranking by raw rating is surface obscure, low vote perfumes. A Bayesian adjustment gives a trustworthy top list.
- Raw Co-occurrence: The raw co-occurrence reflects popularity. Lift and PMI reveal the pairing that actually matter, and also expose a data quality problem with synonym labels.
- Chord diagram: The diagram can be improved, the lift-based pair list exported by the analysis notebook is a better edge list than raw counts.

Data and Cleaning:

The file is semicolon-delimited, latin-1 encoded, and uses a decimal comma in ratings. Fields include perfume, brand, country, gender, rating value and count, release year, perfumer, top/middle/base notes and five main accords.

Check	**Result**
Perfumes	24,063 (no duplicate URLs)
Rating range	1.42 to 4.93
Global weighted mean rating	3.997
Missing release year	8.5%
Known perfumer	48.8%
Average notes per perfume	9.8
Average accords per perfume	4.9

Cleaning steps: converted decimal-comma ratings to floats, treated "unknown" perfumers as missing, split note strings into top, middle and base lists, and built a per-perfume set of unique notes and a list of accords.
Bayesian-adjusted rating. To stop low-count perfumes from dominating, ratings are shrunk toward the global mean with a prior strength of 100 pseudo-votes:

$$adjusted = (n \times rating + 100 \times global\ mean) / (n + 100)$$

Chord Diagram (Fragrance.ipynb):

The first notebook builds the data for an interactive chord diagram of note pairings.

- Each perfume becomes a set of notes, and the notebook records where each note sits in the pyramid (top, middle, base).
- The 80 most frequent notes form the wheel, and the 900 strongest pairs are kept (the page can show 100, 250, 500 or 900).
- Notes are assigned to nine families (Citrus, Floral, Fruity, Green and herbal, Spice, Woods and moss, Amber and sweet, Musk and clean, Resin and leather) using keyword rules plus hand overrides for notes the rules misclassify.
- Output is a self-contained HTML page, scent_chords.html, which the notebook then displays in an iframe and opens in the browser.

Improved chord diagram (v2)

The chord diagram was rebuilt from note_pairs_lift.csv, the lift-ranked pair list exported by the analysis notebook, and published as an interactive page. The file scent_chords_v2.html is on the same page.

Aspect	**Original**	**Improved (v2)**
Pair ranking	Strongest raw co-occurrence, which favors popular notes such as musk	Lift (at least 1.3, with at least 40 co-occurrences), with a toggle back to raw count
Note names	Variant labels kept separate	Synonyms merged (vanille and vanilla, virginia cedar and cedar, white musk and musk, hiacynth and hyacinth, regional roses and lemons)
Note colors	Nine hand-written keyword families	Five Louvain communities detected from the pair network
Notes on the wheel	80	55
Pairs shown	Fixed at 100, 250, 500 or 900	Slider from 40 to 320 pairs
Interaction	Pair filter	Hover a note to isolate its ribbons and list its best partners with lift; hover a ribbon for its count and lift

The five communities, named from their most frequent notes:

Community	Notes	Top notes
Classic florals	19	jasmine, rose, lily-of-the-valley, bergamot, ylang-ylang
Fresh and aromatic	14	vetiver, lavender, geranium, cardamom, nutmeg
Warm and resinous	12	patchouli, cinnamon, tonka, leather, benzoin
Woods and chypre	5	cedar, vanilla, oakmoss, lemon, orange blossom
Fruity florals	5	freesia, black currant, raspberry, peony, pear

Caveats. When two variant notes were merged, their counts were summed and their lifts averaged weighted by count, so merged values are approximate. Note size reflects each note's total pair count rather than its true frequency, because fra_cleaned.csv was not available for this rebuild. Recomputing from the raw file would make both exact.

Exploratory Analysis:

The catalog is heavily recent: 11,533 perfumes date from the 2010s and 5,309 from 2020 onward, against only 67 from the 1960s and 171 from the 1970s. Decade comparisons for older eras therefore rest on small samples.

Accords over time, gender and country. The notebook plots the share of perfumes using each of the top ten accords by decade, the accord share difference between women's and men's perfumes, and a country-by-accord heatmap standardized across the eight largest countries so that relative skew stands out rather than overall popularity.

Brands (349 brands with at least 20 perfumes).

View	Leaders
Highest median adjusted rating	Arabian Oud (4.153), Chanel (4.145), Aramis (4.143), Lorenzo Pazzaglia (4.142), Areej Le Dore (4.132)
Most signature (one dominant lead accord)	Vertus (54.5% woody), Elie Saab (50.0% white floral), Roger and Gallet (48.6% citrus)
Most diverse lead accords	Zara (entropy 4.29), Zoologist (4.18), Avon (4.13)

Chanel is notable because it combines a top-tier rating with a very large rating base (about 362,000 ratings). Guerlain is similar, with about 369,000 ratings and a median adjusted rating of 4.107.

Pyramid position. For notes appearing in at least 200 perfumes, the notebook computes the share of appearances in the top, middle and base layers, identifying which notes mainly open a fragrance and which anchor the drydown.

Note Associations: Lift and PMI

Raw co-occurrence mostly reflects popularity. The most frequent pairs (bergamot with musk at 4,455 perfumes, jasmine with musk at 4,282) have lift of only 1.14 to 1.29, barely above chance.

Lift and PMI correct for this:

$$lift(a, b) = P(a, b) / (P(a) \times P(b)), \; and \; PMI = log2(lift)$$

After keeping the 365 notes appearing in at least 50 perfumes and the 6,360 pairs seen together at least 30 times, the most over-represented pairs are:

Pair	**Count**	**Lift**
castoreum + civet	59	24.7
cypriol oil + gurjan balsam	30	23.0
davana + rum	32	15.7
fig + fig leaf	33	15.2
calabrian bergamot + sicilian lemon	30	12.4
amalfi lemon + oak moss	84	11.9
african orange flower + vanille	134	11.3

These read as genuine perfumery idioms: animalic accords, balsamic resins, sweet-honey stacks and chypre-style citrus with moss.

Data-quality finding. The pairs with the lowest lift among common notes are almost all synonym or near-duplicate labels: vanilla/vanille, cedar/virginia cedar and musk/white musk. Perfumers rarely list both, so the labels appear to repel each other. Merging these variants before building the chord diagram would produce cleaner results.

Statistical Tests:

Tests use the 18,958 perfumes with at least 50 ratings, and non-parametric methods because ratings are skewed.

Rating by gender label. The Kruskal-Wallis test is highly significant (H = 738.8, p = 3.8e-161). Median ratings are 4.05 for men's, 4.00 for unisex and 3.92 for women's perfumes. All pairwise Mann-Whitney tests (Bonferroni corrected) are significant, with the largest effect between women and men (rank-biserial 0.28).

Rating by lead accord. Also significant (H = 1061.9), but the effect size is small (epsilon-squared = 0.059): lead accord explains only about 6% of rank variation in rating.

Popularity versus rating. The Spearman correlation between rating count and rating is 0.037. It is statistically significant given the large sample but practically negligible, so popular perfumes are not noticeably better or worse rated.

Raw versus adjusted rankings.

Ranking	**Top results**
Raw rating	Narmar Extrait de Parfum (4.93 from 30 votes), Aoud No 1 (4.86 from 28), Khaox (4.85 from 26)
Adjusted rating	Valentino Uomo Intense 2021 (4.617), Le Male Le Parfum (4.596), The Most Wanted Parfum (4.575)

The raw leaders all have under 50 ratings. The adjusted list is dominated by well-known perfumes with hundreds to thousands of votes.

Content-Based Recommender:

Each perfume is represented as a bag of tokens: its notes, its accords (lead accords counted twice) and position-tagged notes. TF-IDF weighting (2,677 features, minimum document frequency 5) down-weights ubiquitous notes such as musk and bergamot, and cosine similarity ranks neighbours.

- Sauvage (Dior) returns Sauvage Very Cool Spray (0.864), I Am The King (0.842) and Vodka Wild (0.630), all sharing the fresh spicy, citrus and aromatic profile.
- Aventus (Creed) returns fruity, smoky, leather alternatives, including Supremacy Silver (0.604) and Club De Nuit Intense Man (0.434).

Validation. Across 500 random perfumes, the top-10 neighbours share the query's lead accord 34.8% of the time, against a random baseline of about 7.5%, nearly a fivefold improvement.

Predictive Models:

Predicting rating:

Features (367 in total): multi-hot notes appearing in at least 100 perfumes, accords, gender, release year and top-8 countries. Target: rating, on perfumes with at least 50 ratings, using an 80/20 split.

Model	**R-squared**	**RMSE**
Mean baseline	0.000	0.259
Ridge	0.135	0.241
Histogram Gradient Boosting	0.161	0.237

Both models beat the baseline, but explain only 13 to 16% of variance. This is expected: taste is subjective and rating is influenced by factors the data does not capture, such as price, marketing, and flankers sharing a parent's reputation. In the Ridge model, brand-origin effects are small, with Brazil (+0.123) and UAE (+0.095) the largest country coefficients, and men's labelled perfumes carrying +0.09 relative to women's.

Predicting the gender label

A class-balanced logistic regression on notes and accords reaches 71.9% accuracy.

Class	**Precision**	**Recall**	**F1**
Men	0.636	0.784	0.702
Unisex	0.619	0.539	0.576
Women	0.824	0.812	0.818

Unisex is hardest, as expected, since it overlaps both other classes. The most indicative features match convention: lavender, sage, whiskey, leather and watery notes for men; narcissus, hyacinth, peony, orchid and white rose for women; and vetiver, oud, beeswax and rose varieties for unisex.

Clustering: Data-Driven Scent Families:

Truncated SVD reduces the TF-IDF matrix to 40 components (27.1% of variance), and k-means with k = 10 produces these families:

Cluster	**Size**	**Defining accords**	**Women's share**
0	1,483	iris, violet, powdery, aldehydic	60%
1	1,794	lavender, herbal, fresh spicy, aromatic	8%

2	2,391	almond, vanilla, coconut, nutty	58%
3	3,668	smoky, balsamic, cinnamon, whiskey	20%
4	2,731	tuberose, white floral, yellow floral, animalic	74%
5	1,578	rose, floral, patchouli, musky	75%
6	3,100	tropical, fruity, cherry, caramel	73%
7	3,861	conifer, marine, cannabis, salty	16%
8	2,532	aquatic, fresh, floral, ozonic	not shown
9	925	oud, leather, rose, smoky	not shown

The adjusted Rand index against lead accord is 0.195, so the clusters are related to lead accord but capture something different. They group perfumes by overall note composition rather than a single accord. This supports replacing the hand-written keyword families in the chord diagram with data-driven ones.

Note Network and Communities:

A graph of the 150 most common notes, with edges for pairs having lift of at least 1.5 and at least 40 co-occurrences, yields 147 nodes and 1,181 edges. Louvain community detection finds five groups:

Community	**Size**	**Character**	**Example notes**
0	38	Fruity-floral, fresh	mandarin orange, lily-of-the-valley, peach, freesia, white musk
1	34	Classic warm floral	musk, bergamot, sandalwood, amber, jasmine, vanilla, rose
2	33	Fresh aromatic and woody	cedar, vetiver, lavender, cardamom, geranium, oakmoss
3	30	Oriental and resinous	patchouli, leather, benzoin, oud, incense, saffron
4	12	Citrus and label variants	orange, virginia cedar, vanille, amalfi lemon

Community 4 is largely an artifact of the synonym-label issue from the ‘Note Associations: Lift and PMI” section. The "bridge" notes with the highest betweenness are carnation, peach, green

notes, oakmoss, ylang-ylang, lavender and lily-of-the-valley, which link otherwise separate groups.

Limitations:

- Synonym and variant note names inflate the vocabulary and distort association statistics.
- Ratings are user-generated and reflect fan communities (for example, flankers inheriting a flagship's score).
- Older decades have very small samples.
- Year is missing for 8.5% of perfumes and perfumer for about half.
- Ridge coefficients and associations describe patterns, not causes.
- Model results use one 80/20 split without cross-validation or confidence intervals.

Output Files:

- scent_chords.html: chord diagram page
- note_pairs_lift.csv: lift-ranked note pairs
- perfume_clusters.csv: per-perfume cluster, rating and adjusted rating